

Today we finish talking about FDIA-based attacks. To recap, under this type of attack we can discuss:

- topology attacks.
 - load redistribution attacks.
 - complex, multilevel attacks.
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Specifically, we can describe state preserving attacks. We can see how $a = (H - \bar{H})x$ will lead to $z_a = z + a = Hx + \bar{H}x - Hx = \bar{H}x$, with $z_a = \bar{H}x$. The state thus remains unaffected, and the topology changed to $\bar{H}$. The attacker does not know x , and will use $\hat{x}$ instead of x , but to compute $\hat{x}$ the attacker will need z , as $\hat{x} = (H^TWH)^{-1}H^TWz$. This requires full disclosure resources.

A quick introduction to Load Redistribution attacks: in these kind of attacks, the attack will not alter the power measurements of the generators, as typically power measurements are well-protected. The attacker does not inject false data inside the power measurement of generators, rather he will only target power measurements of consumption and flows on lines.

If we take a look at before the attack and after the attack, we can see that:

$$\begin{aligned} \sum P_G &= \sum P_G^a \\ &= \\ \sum P_{consumed} &= \sum P_{consumed}^a \end{aligned}$$

The sum of the power generated before the attack is equivalent to the power after the attack. The idea is that the attack will not change the power of the generated and consumed power, rather just the distribution.

Let's consider a simple system with three buses (as a complete graph). In lines we might have, before the attack, 10 MW flowing and on another 9 MW, but after the attack the operator might see that on the former line 5 MW are flowing, while on the latter 14 MW. This shows that the sum of the power flowing does not change, rather just the distribution.

Lines might have a safety limit set, like on the latter line we could have a safety limit of 14. The operator would then start some procedures to bring the flow back

to normal levels, as he sees the power flowing too close to the limit (while, in reality, only 5 MW are flowing). We could also have the opposite case, in which the line is overloaded but to the operator it looks like it's completely fine.

Imagine standard FDIA as a basic ingredient of a more complex attack. We could have a load redistribution attack mixed with a topology attack. But now, let's focus on a more complicated situation. Take the real state of the system x , and the estimated state (under attack) $\hat{x}_a$. We then also have physical commands sent by the controller, basing itself on the estimated state, to the system.

Suppose that the goal of the attacker is to cause damage to a specific line in the grid. If we have too much power flowing on a cable, it will explode. Let's set the limit to 14 MW. We see how an attacker can build the attack to trigger a control action that will increase a lot the power flowing on that cable, with the power going above the limit, leading to the explosion of the cable, or the disconnection through a mechanism of protection.

The attacker will falsify data to induce a wrong state of the grid, which would trigger control action (wrong ones) that will be injected to the system, leading to damages. This takes the name of complex multi-level attacks. Here, the FDIA is just a part of the attack.

Let's analyse a specific scheme (slide 16): we have the real system (the grid), including the generation, transmission, distribution and prosumers. We also have sensors sending data back to the SCADA network (both analog and digital measurements). We then have the SE and BDD. After passing the BDD, the estimated $\hat{x}$ is sent to other modules in the system. Let's focus on a specific block, the security assessment. This module is used by the operator to understand how good the estimated state is, considering the limits of the grid. It will tell to other modules to do something about the situation, and typically this is in the form of security constraints, telling other modules to respect a specific constraint. So, the goal is to understand how good is the current state, and if there are problems it gives instructions to other modules, dictating how they should act. These instructions are constraints. Then, data from the security assessment is passed to the SCED, OPF and ED module. Let's start by the ED, the Economic Dispatch, it basically tells every power plant how much power they need to produce. We need to have that $\sum P_G = \sum P_D$. In the grid, we can forecast how much power we need to generate for the next day, and we must decide how to generate it. If we have 10 power plants, and know we need to produce 2 GW, there are different ways to perform this. It's called Economic Dispatch as it's an optimisation problem, and the goal is to minimise the generation costs. We want to generate the power in the

most economical way. The SCED problem, in the other hand, is a version of ED, taking the name of Security Constrained Economic Dispatch. It's basically the ED + security constraints. We have to minimise the total generation costs, but we have also some constraints coming from the security assessment. The output of this is a set point for generators, going to dispatchers that inform generators how much power they need to produce. The problem of ED also determinise the cost of electricity, as we're determining which generators are working and how much power they're generating. We're coupling with electrical markets, and there're specifying attack targeting the markets. The AGC module is the Automatic Generation Control module. Imagine that each generator has a target, like generator 1 having to generate 100 MW, and generator 2 needing to generate 1 GW. AGC is a controller which will slightly adjust the generating power to compensate for fluctuations in the frequency. The ED is also a target for AGC, as each controller knows the nominal set point, and then they can oscillate a bit to account for oscillations. The changes are computed based on a measurement of the frequency. If the frequency is too high for example, we're generating too much and need to decrease the quantity.

We will understand how to have a FDIA targeting the SCED, which will lead to it outputting set points that are wrong, leading to the power on a line being too high or too low. (This is an example).

We start by looking at the SCED problem, and then we see the formulation of the complex attack problem.

So, the goal is a topology load redistribution attack optimised to maximise a given damage criterion (for example, the power flowing on a target line to destroy it). Attacker induces a false state of the grid, which will trigger wrong actions, which will lead to the intended goal.

How does the SCED modules work? The problem is optimally determine generator's set points, to minimise the generation costs. Let's say we have G generators, and define $c_i(\cdot)$ the cost function of generator i .

Each generator has thus a different technology and cost in generating. c_i tells us how much it cost for a generator i to produce the inputted amount of power.

$$\begin{aligned}
& \min_{P_1^g, \dots, P_G^g} \sum_{i=1}^G c_i(P_i^g) \\
& \text{subject to} \\
& \quad P_i^{g,min} \leq P_i^g \leq P_i^{g,max} \quad (\text{local}) \\
& \quad \sum_{i=1}^G P_i^g = \sum_{i=1}^L P_i^d \quad (\text{global}) \\
& \quad P_{ij}^{min} \leq P_{ij} \leq P_{ij}^{max} \quad \forall (i,j) \text{ connected} \\
& \quad V_i^{min} \leq V_i \leq V_i^{max} \quad \forall i \text{ bus of grid}
\end{aligned}$$

P_i^g is the power generated by generator i .

The last two limits are decided by the security assessment module, while the " $\forall (i,j)$ connected" depends on the topology. This problem depends on the estimated state of the grid (both analogue and digital).

The output of this problem is then given by the optimal set points to give to generators, P_i^g , but also the optimal power flow P_{ij}^* . Instead of writing these equations, we write $P^g = P_{ij}^* = SCED(\hat{\mathcal{G}})$.

We also need to know the demand for the problem.

Let's see now the full attack formulation. We're building the attacker problem.

It's a maximisation problem, as the attacker is maximise a given damage function.

$$\begin{aligned}
& \max_a f^{attack}(P_i^{*,g}, P_{ij}^*, P_i^{\oplus,g}, P_{ij}^{\oplus}) \\
& \text{subject to} \\
& \quad (P_i^{*,g}, P_{ij}^*) = SCED(\mathcal{G}) \quad (\text{baseline}) \\
& \quad (P_i^{X,g}, P_{ij}^X) = SCED(\bar{\mathcal{G}}) \\
& \quad (P_i^{\oplus,g}, P_{ij}^{\oplus}) = PF(P_i^{X,g}, \mathcal{G}) \\
& \quad (a,b) = \text{undetectable load redistribution}
\end{aligned}$$

f^{attack} is a mathematical representation of what the attacker aims to damage. We can think of it as the power flowing on a given line, P_{ij} .

A first constraint might be given by the attacker simulating the SCED problem on the real topology ($\mathcal{G}$). Basically, he is simulating what the operator will do in absence of attacks, computing the optimal generation setpoints. This is called the baseline.

Note that P^X indicates the optimal values for $\bar{\mathcal{G}}$, but not for $\mathcal{G}$.

In the second constraint, the operator is simulating the operations of how the operator would do things in the control center. With the second step, the attacker can compute the control actions sent to the grid for the topology $\bar{\mathcal{G}}$.

PF are the power flow equations, modelling the system. They can be used by the

attacker to simulate the effect of wrong actions on the real grid (and that's why we use the real topology).

If we want to maximise P_{ij} , we first need to compute P_{ij} , and so the target function also includes $P_i^{\oplus,g}, P_{ij}^{\oplus}$.

An idea for (a, b) could be given by $a_i = 0 \ \forall i \notin K$, and $c_i = 0 \ \forall i \in Q$, ...

So, we need to find the attack vector that will maximise a given target function, and what we include in these problems is: the baseline of the operation (in absence of an attack), the action taken by the operator (under wrong state assumption), the real actions on the system (simulated), and the ??